

Task 2: WLAN and mobile security plan

A. WLAN vulnerabilities

One vulnerability that Alliah's WLAN is having is the non-mentioning of the type of WLAN encryption methods. Her WLAN could be a target for unauthorized access due to the absence of security protocols. If Alliah's WLAN has out of date/weak encryption methods, it can be a big target for attackers. The second WLAN vulnerability that Alliah has is the access point placement outside the patio. This placement can be vulnerable to war driving, which is a hacking method where an attacker is moving around using hacking software to find vulnerable Wi-Fi networks around them. He could crack into the Wi-Fi network, allowing unauthorized access.

B. Mobile Vulnerabilities

One mobile vulnerability that Alliah could have is with their employees who are representatives that are on the road 80% of the time. Each rep has their own device, their company issues laptops, tablets, and smartphones. This is a vulnerability because employees who are constantly on the road have a risk of losing their devices or being stolen, and it could be a breach in confidentiality for company data if these devices are not secured. Another mobile vulnerability that Alliah could have is with her BYOD policy. BYOD is when employees bring their own device to use within the company, which could be a security risk because personal devices might have a lower level of security than company-owned devices.

C. Mitigation Steps

For Alliah's WLAN vulnerabilities, they should implement WPA3 encryption to their WLAN because it has AES encryption that is used for protecting electronic data. WPA3 also has password authentication protection which is strong enough to not get cracked by unauthorized users. For the WLAN placement, they should conduct a site survey to help find out the best placement to minimize over-extended coverage and interference from the outside. For mobile vulnerabilities, they should add mobile device management software, which allows for control over any device security wise. Through this software, they should implement a mechanism where they're able to wipe the device from its data in case it gets stolen. For BYOD protection, they should implement anti-malware/virus on all devices in case of malware infections, as well as employee training of what to look out for against cyber-attacks.

D. Preventative Measures

Preventative Measures for WLAN include consistent security audits as well as updating security patches as much as possible to mitigate future vulnerabilities. To add on to that, incorporating network segmentation where you separate important systems from regular ones could help lessen the impact of the breach. They could follow The Payment Card Industry Data Security Standard (PCI DSS) as a guideline when doing security monitoring and preventing security breaches. Another preventative measure for mobile security which includes BYOD can be employee training, where they could educate them on potential threats like phishing attacks, and the importance of having unique passwords. This could be outlined by the Federal Trade Commission (FTC) on their "cybersecurity for small businesses" section. When using MDM for

mobile security, they need to also follow SOX (Sarbanes-Oxley Act), this act could make sure to identify, protect and safeguard financial data.

E. BYOD Approach

The best approach to implementing BYOD is to involve SASE (Secure Access Service Edge). This is a service that brings network security like secure web gateways, firewall as a service, and zero trust network access, as well as service capabilities like an SD-WAN. This is a cheap alternative to implementing a more secure way of using BYOD. All of this is accessed and accomplished through the cloud, "It helps you better manage security by extending controls to devices, remote users, and distributed locations anywhere in minutes" (CISCO). Because it's a cloud service, this can be accessed securely anywhere for the Alliah's employees.

F. Acknowledge sources

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